REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	3121
Normal Flows	2928
Detected Attacks	193
Attack Rate	6.18%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.30
Validation Accuracy	80.96%
Attack Precision	0.93
Attack Recall	0.72
Attack F1-Score	0.81

3. Network Protocol Distribution

Protocol	Flow Count
TCP	1878
UDP	1241
OTHER	2

4. Detected Attack History

Time	Source	Destination	Protocol	Confidence
2026-08-25 17:23:44	10.2.0.1:65432	192.168.1.18:51989	TCP	42.68%
2026-08-25 17:23:01	192.168.1.20:34619	239.255.255.250:1900	UDP	77.64%
2026-08-25 17:22:58	192.168.1.18:59878	202.88.211.111:443	UDP	30.02%
2026-08-25 17:22:50	192.168.1.18:63475	4.150.223.103:443	TCP	46.74%
2026-08-25 17:22:49	192.168.1.18:60592	20.199.39.224:443	TCP	33.09%
2026-08-25 17:22:43	10.2.0.1:65432	192.168.1.18:64131	TCP	42.68%
2026-08-25 17:22:11	192.168.1.18:49665	4.2.2.2:53	UDP	85.17%
2026-08-25 17:22:11	192.168.1.18:51802	4.2.2.2:53	UDP	58.99%
2026-08-25 17:22:03	192.168.1.18:63475	4.150.223.103:443	TCP	51.38%
2026-08-25 17:22:01	192.168.1.18:60592	20.199.39.224:443	TCP	33.07%
2026-08-25 17:21:40	150.171.110.84:443	192.168.1.18:53035	TCP	57.39%
2026-08-25 17:21:36	172.64.155.209:443	192.168.1.18:64090	TCP	41.62%
2026-08-25 17:21:35	10.2.0.1:65432	192.168.1.18:56288	TCP	42.68%
2026-08-25 17:21:32	142.251.152.119:443	192.168.1.18:50431	UDP	81.74%
2026-08-25 17:21:08	192.168.1.18:63475	4.150.223.103:443	TCP	72.81%
2026-08-25 17:21:07	192.168.1.18:60592	20.199.39.224:443	TCP	54.95%
2026-08-25 17:20:53	142.251.151.4:443	192.168.1.18:50708	UDP	38.31%
2026-08-25 17:20:45	192.168.1.18:63475	4.150.223.103:443	TCP	45.72%
2026-08-25 17:20:40	150.171.28.11:443	192.168.1.18:60097	TCP	38.02%
2026-08-25 17:20:40	192.168.1.18:60592	20.199.39.224:443	TCP	54.61%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.